

Element C: Presentation and Justification of Solution Design Requirements

People can easily lose things such as wallets, identification cards, phones, etc. from their pockets and need a way to keep things in their pockets because those items fall out of their pockets easily. Anyone who has had items fall out of their pockets would be involved in our project. This problem occurs everywhere in the world, from at home to traversing a big city. People often lose items from their pockets because their pockets are too small, so the items do not fully fit into the pockets.

1. Physical product, making sure we have a product (high priority)
2. Safe to use - no sharp objects or edges - safe to touch (high priority)
3. Does not interfere with electronics (high priority)
4. Under \$15 – per product (high priority)
5. Lightweight – it doesn't pull any pants down (high priority)
6. Longevity – product lasts at least a year (high priority)
7. Convenient size – no more than 10" in any direction (medium priority)
8. Removable it is easy to take on and off (low priority)
9. Waterproof make it nice but not the most important (low priority)
10. Has multiple color options that allows us to have to option to match (low priority)

This will help us reach our solution and meet our stakeholders' needs. This will give everybody who loses items a way to retrieve them. We talked with Mr. H, who is a math teacher here at our school. who looked over our design requirements, and gave use amazing feedback on what we need to change, and he liked x

These requirements add to the design toward inexpensive, durable materials such as polymers and silicone, ensuring the product is lightweight and safe for daily use. By strictly limiting cost, size, and weight, you guarantee a solution that stakeholders can realistically manufacture while ensuring high user adoption through comfort and electronics compatibility. This helps keep medical equipment, phones, safe from magnets.